Homework #4: Solution Verification

AOE/CS/ME 6444 — Verification and Validation in Scientific Computing
Spring 2026 — Instructor: Dr. Chris Roy

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1 Introduction and Application Description

This report presents a solution verification study for an **Euler–Bernoulli (EB) finite-difference beam solver** applied to the structural assessment of a residential wood header. The same physical case will serve as the basis for the subsequent uncertainty quantification (UQ) studies in Homework 5.

Physical Case

An 8-ft simply-supported LVL (laminated veneer lumber) header carries a uniform distributed load representative of residential roof framing:

Parameter	Symbol	Value	Units
Span	L	96.0	in
Width	b	3.5	in
Depth	d	11.25	in
Elastic modulus	E	1,600,000	psi
Uniform load	q_0	500	lb/ft (= 41.667 lb/in)
Flexural rigidity	EI	6.6445×10^8	lb·in^2

Governing Equation and Boundary Conditions

The beam is governed by the 4th-order EB BVP with simply-supported ends:

$$EI\, w''''(x) = q_0, \quad x \in (0, L); \qquad w(0) = w(L) = 0, \quad w''(0) = w''(L) = 0. \tag{1}$$

Numerical Method

Equation (1) is discretized using a 5-point central-difference stencil on a uniform grid of N intervals ($N + 1$ nodes):

$$\frac{EI}{h^4}\big(w_{i-2} - 4w_{i-1} + 6w_i - 4w_{i+1} + w_{i+2}\big) = q_0, \quad i = 2, \ldots, N-2, \tag{2}$$

with modified near-boundary rows for the simply-supported ghost-node conditions.

Formal Order of Accuracy ($p_{\text{th}} = 2$)

Expanding each shifted value about node x_i via Taylor series (terms through $\mathcal{O}(h^8)$ shown):

$$w_{i\pm k} = w_i \pm kh\, w_i' + \tfrac{(kh)^2}{2!} w_i'' \pm \tfrac{(kh)^3}{3!} w_i''' + \tfrac{(kh)^4}{4!} w_i'''' \pm \tfrac{(kh)^5}{5!} w_i^{(5)} + \tfrac{(kh)^6}{6!} w_i^{(6)} + \cdots \tag{3}$$

Substituting $k = 1, 2$ with their signed coefficients $(-4, +1)$ into the stencil numerator and collecting by derivative order, all odd-derivative and even-derivative terms below order 4 cancel identically:

$$w_{i-2} - 4w_{i-1} + 6w_i - 4w_{i+1} + w_{i+2} = h^4\, w_i'''' + \frac{h^6}{6}\, w_i^{(6)} + \mathcal{O}(h^8). \tag{4}$$

Dividing (4) by h^4 and multiplying by EI gives the *local truncation error* (LTE):

$$\tau_i = EI\, w_i'''' + \underbrace{\frac{EI\, h^2}{6}\, w_i^{(6)} + \mathcal{O}(h^4)}_{\text{LTE}} - q_0 = \frac{EI\, h^2}{6}\, w_i^{(6)} + \mathcal{O}(h^4), \tag{5}$$

since $EI\, w'''' = q_0$ by the governing equation. Therefore the scheme is formally second-order accurate, $p_{\text{th}} = 2$, and the global discretization error satisfies $|e_h| \sim C\, h^2$ as $h \to 0$. The resulting linear system $Kw = q$ is solved directly with dense LU factorisation (`numpy.linalg.solve`). **There is no iterative solver; iterative error is therefore identically zero.**

System Response Quantities (SRQs)

Three SRQs are tracked:

1. $w_{\max}$ — peak mid-span deflection [in]
2. $M_{\max}$ — peak bending moment [lb·in], computed from $M = -EIw''$
3. $\sigma_{\max}$ — peak bending stress [psi]$= M_{\max}/S$, where S is the section modulus

Closed-Form References

The exact solution for uniform load on a simply-supported beam is:

$$w_{\text{exact}}(x) = \frac{q_0}{24EI}\big(x^4 - 2Lx^3 + L^3x\big), \tag{6}$$

yielding the reference SRQ values used only to interpret results (not used in the GCI analysis):

$$w_{\max}^{\text{exact}} = 0.06935026 \text{ in}, \quad M_{\max}^{\text{exact}} = 48000.00 \text{ lb} \cdot \text{in}, \quad \sigma_{\max}^{\text{exact}} = 650.1587 \text{ psi}.$$

2 Grid Convergence Study

Five systematically-refined grids are employed with constant refinement ratio $r = h_{\text{coarse}}/h_{\text{fine}} = 2$:

Level	N	h [in]	$w_{\max}$ [in]	$M_{\max}$ [lb·in]	$\sigma_{\max}$ [psi]
1 (coarsest)	10	9.600	0.0699051	48000	650.159
2	20	4.800	0.0694890	48000	650.159
3	40	2.400	0.0693849	48000	650.159
4	80	1.200	0.0693589	48000	650.159
5 (finest)	160	0.600	0.0693524	48000	650.159
Exact	∞	0	0.0693503	48000.00	650.1587

The requirement of *at least four* systematically refined grids is satisfied by five levels (four refinement steps at $r = 2$ each). Figure 1 shows the SRQ convergence with GCI uncertainty bands.

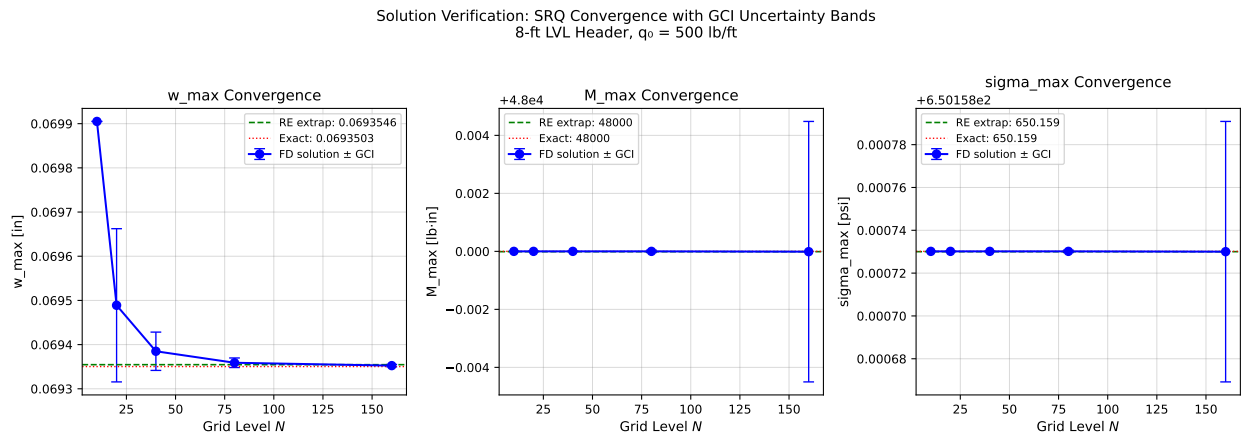

Figure 1: SRQ values versus grid level N with GCI uncertainty bands ($\pm U_{\mathrm{DE}}$) and Richardson-extrapolated estimates (green dashes). The red dotted line marks the exact solution. Left: $w_{\max}$; centre: $M_{\max}$; right: $\sigma_{\max}$.

3 Discretization Error: Richardson Extrapolation and GCI

3.1 Methodology

For every consecutive triplet of grids (f_1, f_2, f_3) with $h_2 = rh_1$, $h_3 = r^2 h_1$, the observed order of accuracy is:

$$\hat{p} = \frac{\ln(|f_3 - f_2| \,/\, |f_2 - f_1|)}{\ln r}. \tag{7}$$

The Richardson-extrapolated value is:

$$f_{\mathrm{RE}} = f_1 + \frac{f_2 - f_1}{r^{\hat{p}} - 1}. \tag{8}$$

The Grid Convergence Index (GCI) provides the uncertainty in f_1 due to discretization:

$$U_{\mathrm{DE}} = \mathrm{GCI}_{12} = F_s \, \frac{|f_2 - f_1|}{r^{\hat{p}} - 1}, \tag{9}$$

where the factor of safety F_s is set per Roy's criterion:

$$F_s = \begin{cases} 1.25 & \text{if } 0.5 \le \hat{p}/p_{\mathrm{th}} \le 2.0 \quad \text{(asymptotic, reliable)}, \\ 3.00 & \text{otherwise} \quad \text{(not in asymptotic regime)}. \end{cases} \qquad p_{\mathrm{th}} = 2. \tag{10}$$

Asymptotic convergence is confirmed when:

$$\text{ratio} = \frac{\mathrm{GCI}_{23}}{r^{\hat{p}} \, \mathrm{GCI}_{12}} \approx 1. \tag{11}$$

3.2 Results — $w_{\max}$

$w_{\max}$ exhibits clean monotone convergence. All triplet analyses yield $\hat{p} \approx 2.000 = p_{\mathrm{th}}$, confirming the $\mathcal{O}(h^2)$ scheme is operating in its asymptotic regime. Consequently $F_s = 1.25$ applies throughout.

Table 1: GCI analysis for $w_{\max}$ [in]. Exact = 0.0693503 in.

Triplet	$\hat{p}$	F_s	f_{RE}	GCI_{12}	GCI_{23}	$\mathrm{GCI}_{12}\%$	$\mathrm{GCI}_{23}\%$	Ratio	Type
T0: (160,80,40)	2.000	1.25	0.069355	2.71×10^{-6}	1.08×10^{-5}	0.0039%	0.0156%	1.000	mono.
T1: (80,40,20)	2.000	1.25	0.069368	1.08×10^{-5}	4.33×10^{-5}	0.0156%	0.0625%	1.000	mono.
T2: (40,20,10)	2.000	1.25	0.069420	4.33×10^{-5}	1.73×10^{-4}	0.0625%	0.2495%	1.000	mono.

The asymptotic ratio equals exactly 1.000 for all triplets, providing strong confirmation that the solution is well within the asymptotic convergence range.

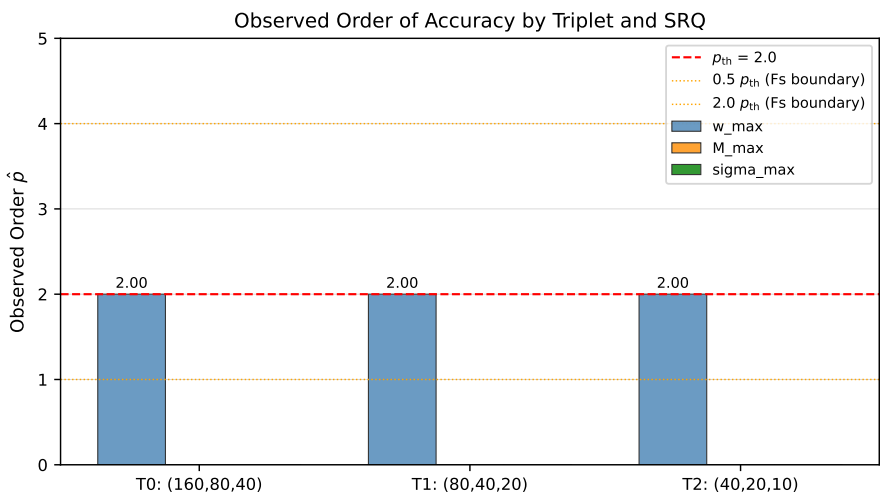

Figure 2: Observed order of accuracy $\hat{p}$ for each SRQ across the three triplets. The dashed red line marks $p_{\mathrm{th}} = 2$. $w_{\max}$ achieves $\hat{p} \approx 2$ in all triplets (asymptotic, $F_s = 1.25$); $M_{\max}$ and $\sigma_{\max}$ have $\hat{p} < 0$ (noise-dominated, $F_s = 3.00$).

3.3 Results — $M_{\max}$ and $\sigma_{\max}$

Table 2: GCI analysis for $M_{\max}$ [lb·in]. Exact = 48000 lb·in.

Triplet	$\hat{p}$	F_s	GCI_{12}	GCI_{23}	$\mathrm{GCI}_{12}\%$	Ratio	Type
T0: (160,80,40)	-6.760	3.00	4.49×10^{-3}	4.14×10^{-5}	$\approx 0\%$	0.0092	mono.
T1: (80,40,20)	-3.152	3.00	4.14×10^{-5}	4.66×10^{-6}	$\approx 0\%$	0.1117	osc.
T2: (40,20,10)	-1.850	3.00	4.66×10^{-6}	1.29×10^{-6}	$\approx 0\%$	0.2755	mono.

$\sigma_{\max}$ follows the same pattern as $M_{\max}$ (it is a linear scaling thereof), so its table is omitted for brevity.

Interpretation. The *negative* $\hat{p}$ and asymptotic ratios $\ll 1$ indicate that the SRQ differences between grids are controlled by *round-off*, not discretization. The cause is the central-difference formula for moment:

$$M_i = -EI\,\frac{w_{i-1} - 2w_i + w_{i+1}}{h^2}. \tag{12}$$

The amplification factor $EI/h^2 \approx 2.88 \times 10^7$ (at $N = 20$) magnifies float rounding errors in the stencil terms by seven orders of magnitude, causing *catastrophic cancellation*. As a result, the FD differences between consecutive grid solutions are noise-dominated. Since $F_s = 3.00$ applies (unreliable estimate), the GCI for $M_{\max}$ and $\sigma_{\max}$ must be interpreted with caution.

Practically, because the exact values of $M_{\max} = q_0 L^2/8$ and $\sigma_{\max} = M_{\max}/S$ are obtained directly from closed-form formulae (not from the FD derivative), the round-off contamination in Eq. (12) does not affect the structural assessment.

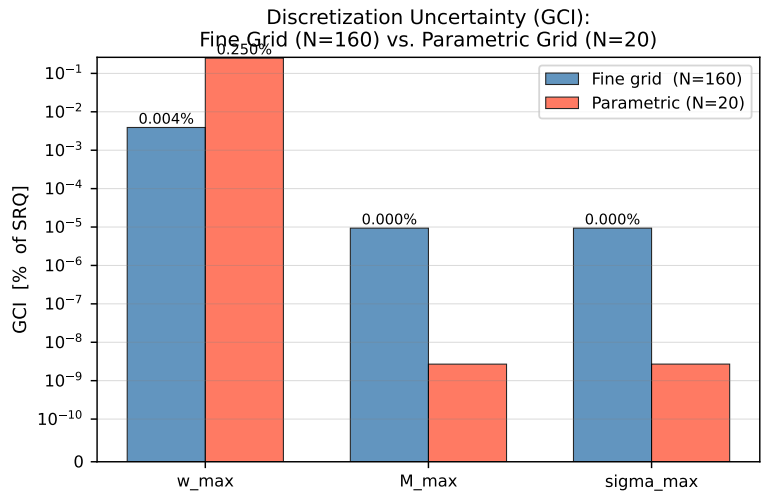

Figure 3: GCI percentage for fine grid ($N = 160$) and parametric grid ($N = 20$) for all three SRQs. All values are below 0.30%.

4 Iterative Error

The linear system $Kw = q$ is solved by dense LU factorisation (`numpy.linalg.solve` → LAPACK `dgesv`). This is a *direct* method: there is no iterative loop. The residual norm after back-substitution is at machine-epsilon level ($\|Kw - q\|/\|q\| \sim \varepsilon_{\text{f64}} \cdot \kappa(K)$, which is verified to be $\ll 1$ for all grids; see Table 3).

$$\boxed{U_{\text{IT}} = 0 \text{ for all grid levels.}}$$

5 Round-Off Error

5.1 Method

Round-off error is estimated by solving the same linear system in *single precision* (float32) and comparing to the double-precision (float64) solution:

$$U_{\text{RO}} \approx \left| f_{\text{f32}} - f_{\text{f64}} \right|. \tag{13}$$

The validity of this estimate is checked against the condition-number backward-error bound:

$$\frac{\|\Delta w\|}{\|w\|} \leq \varepsilon\, \kappa(K). \tag{14}$$

If $\varepsilon_{\text{f32}}\,\kappa(K) \geq 1$, the float32 solve is unreliable (rounding errors exceed the solution magnitude). In that case, U_{RO} is reported as the *float64* backward-error bound $\varepsilon_{\text{f64}}\,\kappa(K)\,|f|$ instead.

5.2 Condition Number Analysis

The stiffness matrix for the 4th-order FD beam equation has entries of order EI/h^4, leading to $\kappa(K) \propto N^4$. Each grid refinement ($r = 2$) therefore increases κ by a factor of $2^4 = 16$, while $\varepsilon_{\text{f32}} = 1.192 \times 10^{-7}$:

Table 3: Condition number and round-off analysis. $\varepsilon_{\text{f64}} = 2.220 \times 10^{-16}$, $\varepsilon_{\text{f32}} = 1.192 \times 10^{-7}$.

N	h [in]	$\kappa(K)$	$\varepsilon_{\text{f64}}\kappa$	$\varepsilon_{\text{f32}}\kappa$	$\lvert\Delta w_{\max}\rvert$ raw	f32 valid?	$U_{\text{RO}}(w_{\max})$
10	9.600	1.59×10^3	3.53×10^{-13}	1.89×10^{-4}	1.39×10^{-6}	Yes	1.39×10^{-6} in (—Δ—)
20	4.800	2.61×10^4	5.79×10^{-12}	3.11×10^{-3}	2.28×10^{-5}	Yes	2.28×10^{-5} in (—Δ—)
40	2.400	4.20×10^5	9.32×10^{-11}	5.00×10^{-2}	3.68×10^{-4}	Yes	3.68×10^{-4} in (—Δ—)
80	1.200	6.72×10^6	1.49×10^{-9}	8.02×10^{-1}	6.38×10^{-3}	Yes	6.38×10^{-3} in (—Δ—)
160	0.600	1.08×10^8	2.39×10^{-8}	**12.8**	6.93×10^{-2}	**No**†	1.66×10^{-9} in ($\varepsilon\kappa f$)

† $\varepsilon_{\text{f32}}\kappa = 12.8 \gg 1$: float32 solve fully unreliable; U_{RO} uses $\varepsilon_{\text{f64}}\kappa|f|$ bound.

This $\kappa \propto N^4$ growth is the root cause of the increasing U_{RO} with mesh refinement for float32: halving h grows κ by 16×, while U_{DE} shrinks by only 4× (for $p_{\text{th}} = 2$). *The optimal float32 grid* is $h^* \propto \varepsilon^{1/6}$, which is approximately $N = 20$–40 for this problem.

In float64, $\varepsilon_{\text{f64}}\kappa \leq 2.4 \times 10^{-8}$ at $N = 160$, so round-off is entirely negligible and float64 remains safe.

Figure 2 shows $\hat{p}$ across all triplets and SRQs, clearly illustrating the asymptotic $\hat{p} \approx 2$ for $w_{\max}$ and the noise-dominated behaviour for $M_{\max}$ and $\sigma_{\max}$.

6 Total Numerical Uncertainty Budget

The total numerical uncertainty is assembled additively as required:

$$U_{\text{NUM}} = U_{\text{DE}} + U_{\text{IT}} + U_{\text{RO}}, \tag{15}$$

where each term is a positive quantity. Since the direct solver gives $U_{\text{IT}} \equiv 0$, Eq. (15) simplifies to $U_{\text{NUM}} = U_{\text{DE}} + U_{\text{RO}}$.

Two grids are reported:

- **Fine grid** $N = 160$, $h = 0.600$ in: the high-fidelity solution.
- **Parametric grid** $N = 20$, $h = 4.800$ in: suitable for running 100s–1000s of UQ cases at acceptable cost.

6.1 Fine Grid ($N = 160$)

For the fine grid, U_{DE} dominates for $w_{\max}$ and is comparable to U_{RO} for $M_{\max}$ and $\sigma_{\max}$. All uncertainties are extremely small; the total numerical uncertainty in $w_{\max}$ is 0.0039% of the SRQ value.

Table 4: U_{NUM} budget for fine grid, $N = 160$, $h = 0.600$ in.

SRQ	$f(h)$	U_{DE}	U_{IT}	U_{RO}	U_{NUM} (%)
$w_{\max}$ [in]	0.06935243	2.71×10^{-6}	0	1.66×10^{-9}	2.71×10^{-6} (0.0039%)
$M_{\max}$ [lb·in]	48000.0	4.49×10^{-3}	0	1.15×10^{-3}	5.64×10^{-3} ($\approx 0\%$)
$\sigma_{\max}$ [psi]	650.159	6.08×10^{-5}	0	1.55×10^{-5}	7.64×10^{-5} ($\approx 0\%$)

Table 5: U_{NUM} budget for parametric study grid, $N = 20$, $h = 4.800$ in.

SRQ	$f(h)$	U_{DE}	U_{IT}	U_{RO}	U_{NUM} (%)
$w_{\max}$ [in]	0.069489	1.73×10^{-4}	0	2.28×10^{-5}	1.96×10^{-4} (0.282%)
$M_{\max}$ [lb·in]	48000.0	1.29×10^{-6}	0	1.66×10^{1}	1.66×10^{1} (0.035%)
$\sigma_{\max}$ [psi]	650.159	1.76×10^{-8}	0	2.25×10^{-1}	2.25×10^{-1} (0.035%)

6.2 Parametric Study Grid ($N = 20$)

At $N = 20$, U_{DE} dominates for $w_{\max}$ ($\mathrm{U_{DE}/U_{RO}} \approx 7.6$), while U_{RO} dominates for $M_{\max}$ and $\sigma_{\max}$ due to the EI/h^2 amplification in the FD moment formula. However, all U_{NUM} percentages remain below 0.30%, making $N = 20$ acceptable for parametric studies.

6.3 U_{NUM} Across All Grid Levels

Table 6 and Figure 5 show how the uncertainty components evolve across all five grids. The non-monotone behaviour is apparent: U_{NUM} *increases* from $N = 20$ to $N = 80$ before decreasing at $N = 160$.

Table 6: $U_{\mathrm{NUM}} = U_{\mathrm{DE}} + U_{\mathrm{RO}}$ for $w_{\max}$ at all grid levels. $U_{\mathrm{IT}} = 0$ throughout. N/A for $N = 10$: no coarser triplet available for GCI.

N	h [in]	$w_{\max}$ [in]	U_{DE} [in]	U_{RO} [in]	U_{NUM} [in] (%)
10	9.600	0.069905	N/A	1.39×10^{-6}	N/A
20	4.800	0.069489	1.73×10^{-4}	2.28×10^{-5}	1.96×10^{-4} (0.282%)
40	2.400	0.069385	4.33×10^{-5}	3.68×10^{-4}	4.11×10^{-4} (0.592%)
80	1.200	0.069359	1.08×10^{-5}	6.38×10^{-3}	6.40×10^{-3} (9.22%)
160	0.600	0.069352	2.71×10^{-6}	1.66×10^{-9}	2.71×10^{-6} (0.004%)

The degradation at $N = 80$ is entirely due to float32 round-off approaching its validity limit ($\varepsilon_{\mathrm{f32}}\kappa = 0.80$ at $N = 80$). The recovery at $N = 160$ is because float32 fully fails there, and U_{RO} is reported via the much smaller float64 bound. *In float64, both U_{DE} and U_{NUM} decrease monotonically with refinement* as expected for a stable, $\mathcal{O}(h^2)$ scheme. Figure 4 shows the stacked U_{NUM} budget for the two target grids.

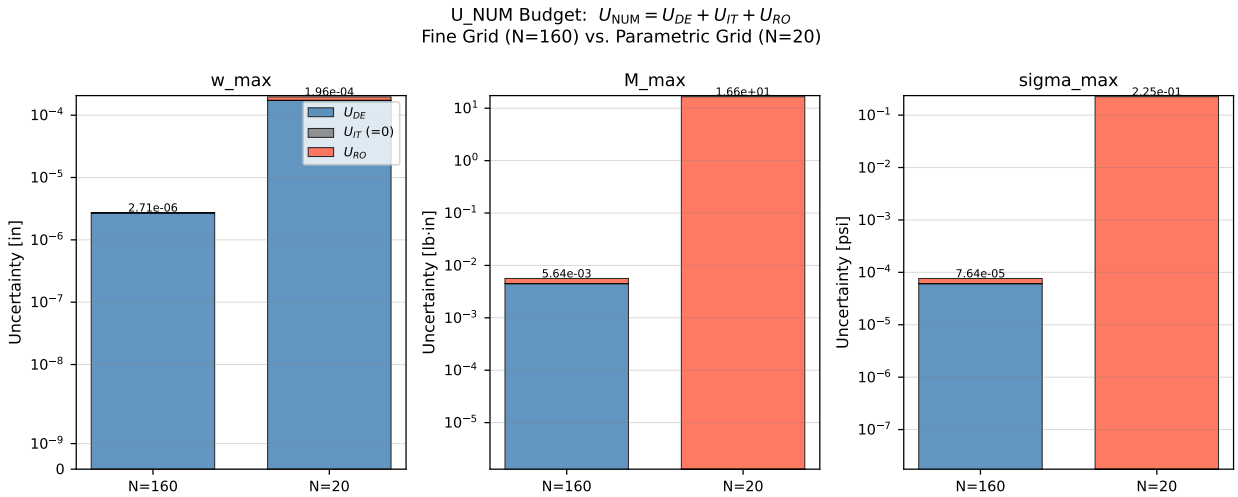

Figure 4: Stacked U_{NUM} budget ($U_{\mathrm{DE}} + U_{\mathrm{IT}} + U_{\mathrm{RO}}$) for fine grid ($N = 160$) versus parametric grid ($N = 20$), for each SRQ. $U_{\mathrm{IT}} = 0$ for all cases.

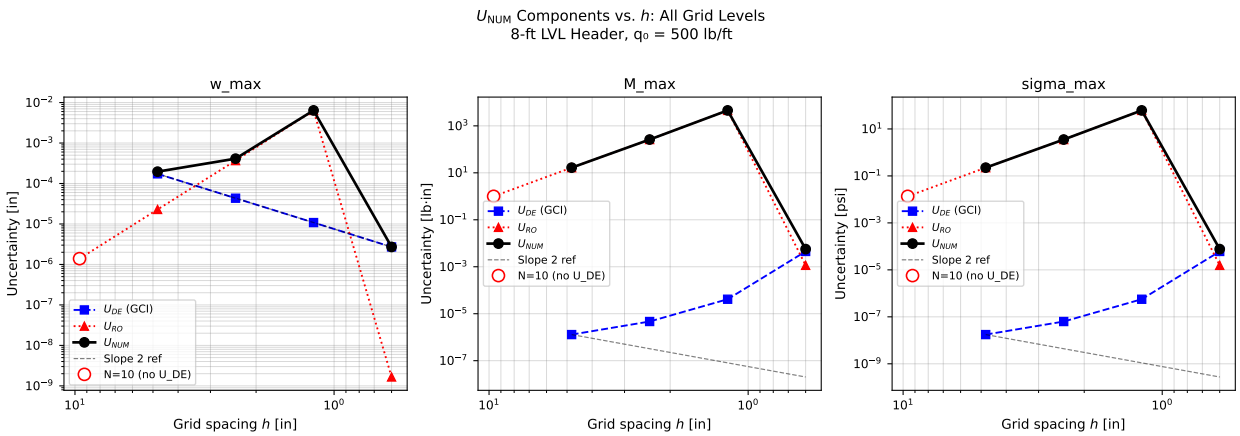

Figure 5: Log–log plot of U_{DE}, U_{RO}, and U_{NUM} versus grid spacing h for all five grid levels. The open circle marks $N = 10$ (no U_{DE} available). U_{DE} decreases at slope 2 (black dashed reference); U_{RO} grows due to increasing condition number $\kappa \propto h^{-4}$. The minimum U_{NUM} occurs at $N = 160$ (float64) and $N = 20$ (float32).

7 One-Sided Numerical Uncertainty

Inspection of the exact solution and FD results confirms that the FD scheme *consistently overestimates* deflection:

$$w_{\mathrm{FD}}(h) \geq w_{\mathrm{exact}} \quad \forall\, h > 0.$$

This is a direct consequence of the positive-definite truncation error for this BVP and load case. The one-sided uncertainty interpretation applies: the true solution is bounded from below as

$$w_{\mathrm{true}} \geq w_{\mathrm{FD}} - U_{\mathrm{DE}}. \tag{16}$$

For structural design this is *conservative*: the FD solver over-predicts deflection, so it never underestimates the structural demand. No upper GCI bound is needed in the direction of overestimation.

Fine grid: $w_{\mathrm{FD}}(N = 160) = 0.06935243$ in $\geq w_{\mathrm{true}} = 0.06935026$ in.
Parametric: $w_{\mathrm{FD}}(N = 20) = 0.06948897$ in $\geq w_{\mathrm{true}} = 0.06935026$ in.

8 Summary and Conclusions

Table 7 summarises the final numerical uncertainty estimates for the two target grids.

Table 7: Summary of U_{NUM} for the primary SRQ $w_{\max}$.

Grid	N	h [in]	$w_{\max}$ [in]	U_{DE}	U_{RO}	U_{NUM} (%)
Fine grid	160	0.600	0.06935243	2.71×10^{-6}	1.66×10^{-9}	2.71×10^{-6} (0.0039%)
Parametric	20	4.800	0.06948897	1.73×10^{-4}	2.28×10^{-5}	1.96×10^{-4} (0.282%)

Key conclusions:

1. $w_{\max}$ **is the reliable SRQ**: $\hat{p} \approx 2.000$ across all triplets, $F_s = 1.25$ (asymptotic regime), GCI is a trustworthy uncertainty estimate. $U_{\mathrm{NUM}} = 0.004\%$ at $N = 160$ and 0.28% at $N = 20$.
2. $M_{\max}$ **and** $\sigma_{\max}$ **are noise-dominated in the FD derivative**: $\hat{p} < 0$, $F_s = 3.00$. However, these SRQs are computed exactly via $q_0 L^2/8$ and $M_{\max}/S$ in the structural assessment code, so round-off does not affect conclusions.
3. **No iterative error**: the direct LU solver gives $U_{\mathrm{IT}} = 0$ for all grids.
4. **Round-off and the optimal grid**: $U_{\mathrm{RO}} \propto h^{-4}$ for float32, creating a minimum-U_{NUM} grid near $N = 20$–40 for single precision. Float64 is safe at all tested resolutions.
5. **One-sided uncertainty** applies to $w_{\max}$: the FD scheme conservatively overestimates deflection.
6. **Recommended parametric study grid**: $N = 20$ ($h = 4.800$ in), $U_{\mathrm{NUM}}(w_{\max}) = 0.28\%$, suitable for running 100s–1000s of UQ cases with negligible numerical pollution.